

Smart Soil Monitoring System With Crop and Fertilizer Recommendation Features

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Abstract—This paper represents the design of a portable and IoT based device for monitoring nitrogen, phosphorus, potassium, pH, and moisture of soil, along with surrounding temperature and humidity, for crops and fertilizer recommendation. The idea and execution of a wireless, portable system for assessing soil quality, recommending the optimal amount of fertilizer, and suggesting appropriate crops depending on the soil conditions are the main goals of this research project. The system is designed to have a user-friendly interface. It uses a cloud-based platform for analyzing data and implementation of machine learning models. Multiple classification-based machine learning algorithms are compared to gain better accuracy of the system. The results of the system are validated using different samples of both indoor and outdoor environments. This cost-effective design, which includes monitoring of seven parameters necessary for crop cultivation along with crop recommendation feature and fertilizer suggestion, makes it desirable for agroindustry.

Keywords— *IoT; soil monitoring; agro technology; fertilizer recommendation system; machine learning*

I. INTRODUCTION

According to FAO, 1.23 billion people are employed in the agrifood system, and 857 million people connected to the agrifood system work in primary agricultural production [1]. However, in emerging, undeveloped nations in Asia and Africa, farmers possess the necessary skills for cultivation but lacks the information regarding the most suitable crop for a particular soil type. The majority of the time, farmers plant a certain crop in response to market demand, but nobody is aware of the level of nutrients in the soil, which might have an impact on optimal production.

Soil contains several micronutrients and macronutrients. Macronutrients are present in significant amounts which are nitrogen, phosphorus, potassium, calcium, magnesium, and sulfur. Out of the mentioned macronutrients, only the consecutive first three nutrients are highly essential and the main factors for crop production. But it was observed that the overall all nitrogen and potassium balance is negative along

with phosphorus balance being near to zero in Bangladesh. Poor farming has always been an issue but using fertilizer without any calculated measurement has fasten the soil degradation [2]. Thus, soil quality degradation and loss of nutrients present within the soil may lead to crop failure and malnutrition, reducing both the quality and quantity of food available to the rapidly growing population [3].

In this paper, a compact portable system capable of collecting soil parameters and suggesting suitable crops for cultivation is proposed. Moreover, this device can also be used for regular monitoring of the soil. In order to identify the specifications for this proposed system, the authors collaborated with both farmers and agriculture experts. Significant insights from the specialists facilitated the identification of the system requirements which differentiates this system from traditional soil monitoring systems. The proposed system carries more features than the conventional devices on the market due to the integration of crop recommendation along with fertilizer suggestions. Consequently, farmers are equipped with the technology to make logical decisions on crop selection, based on machine learning.

Subsequent section II discusses various types of relevant work conducted by other researchers. The methodology, as well as the system design, are discussed in section III. Section IV presents the implementation of the proposed monitoring system. The forthcoming section V of the paper includes the outcomes and observations of the system, as well as the discoveries derived from the analysis. Lastly, section VI concludes the article by combining key facts and implications and suggesting potential areas for future research.

II. LITERATURE REVIEW

Monitoring soil quality, requiring different soil parameter data collection sensors work simultaneously. If the soil data collecting sensors are autonomous and can communicate with each other, then it is referred to as a Wireless Sensor Network (WSN). Madhumati et al employed several sensors in order to propose a real time soil monitoring system [4]. To measure the

quantity of nitrogen, phosphorus, and potassium in soil, NPK tester was used. Along with that, an Arduino microcontroller was used for data collection and Node MCU was utilized for sending data to a web service. Madhura et al proposed a system architecture containing multiple sensor nodes using microcontroller for data collection and incorporated a raspberry pi as a gateway node for sending the collected data to cloud based system [5]. For real time data visualization and data storage Kamelia et al used thinkspeak [6]. A different approach is proposed by Pallevada et al for collecting NPK and pH data. There, a TCS3200 Sensor is used which detects the color change from the soil testing capsules being mixed with the soil [7]. A significant development can be noticed in the IoT system, due to which it is easier to get access to any information or control any system from anywhere. Pyingkodi et al proposed a similar methodology using a GSM module embedded in to the system, which transmits any warning or suggestion through SMS to the user [8]. An image processing-based approach is proposed by Bolla et al where, the image data and moisture are analyzed for soil pH detection. A variety of crops is recommended according to pH amount is presented in the research work [9]. According to Senthilkumar Mathi et. al ML models can predict the demand of different crops and suggest optimal planting times and location [10]. ML can be applied in precision farming by utilizing satellite imagery, drone footage, and soil sensors to create detailed maps of crop growth, nutrient levels, and moisture content. This data can be utilized for fertilizer applications to irrigate specific areas of a field [11].

All such research and projects were done, but there still remains a gap which refrains technology to meet all the requirement for developing a smart device for farmers, which can collect soil parameters to assess land quality, providing fertilizer recommendations to improve land quality, predicting the best suited crop for the land considering current soil health and suggests required fertilizer for different crops. One specific concept is usually investigated for each research project. That is the reason of coming up with a comprehensive system that would help farmers produce the best results possible off their land.

III. METHODOLOGY AND SYSTEM DESIGN

The essential parameters for building such a system include nitrogen, phosphorus, potassium, pH level, and moisture presence in the soil. In addition to that, environmental temperature and humidity are also necessary. The proposed system's block diagram is presented in Fig. 1, which is an IoT-enabled system that can monitor the soil parameters of land in real time along with crop selection and providing fertilizer recommendation based on current soil parameters. For measuring primary nutrients in soil such as nitrogen, phosphorus, and potassium, an NPK sensor is used, which will be inserted into the soil for data collection. This sensor operates in accordance with the proportionality of the ion quantity present in soil to the signal generated by the electrodes. Using a probe-based pH sensor, the system measures the pH level of the soil. Soil pH data is crucial for evaluating the suitability of a crop for a specific land area.

Both the NPK sensor and pH sensor are based on the Modbus communication protocol. This protocol is used to get the readings of NPK and soil pH sensors to the microcontroller of the proposed system. With the help of the FC-28 sensor, soil moisture data is acquired by estimating the soil electric conductivity between the two probes of the sensor. DHT 11 is a two-in-one packaged sensor capable of measuring the temperature and humidity of the surrounding environment. Due to the implementation of IoT in the system, the ESP 32 is used as the processing unit. Meanwhile, the decision-

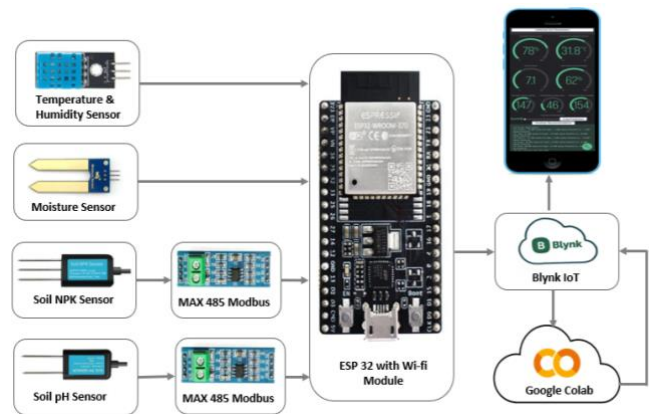


Fig. 1. Block diagram of proposed system

making system for crop selection was implemented using cloud-based machine learning.

For predicting the crop names using the obtained soil data, it is important to select a dataset that is highly compatible with the data collected by the proposed system. The system is trained using the crop recommendation dataset, which consists of seven parameters, each properly labelled with the corresponding crop names [12]. The dataset is classified into multiple categories, and supervised machine learning methods that depend on classification are utilized for this sort of data. For classification-based data, Logistic Regression, Decision Trees, Random Forest, Support Vector Machine (SVM) are the most suitable ML models. The collected data are implied in the machine learning model and based on that a decision is made for which crop is suitable for the collected soil parameters.

Various commercial cloud computing platforms are available which are paid, but Blynk IoT and Google Colab, are employed for the purpose of prototyping. Blynk IoT is renowned for its easy-to-use interface, so this is utilized at the user end. The simple interface of this app has an easy log-in facility, and the farmers can install it in their smart phones of any specifications. In Blynk IoT app, the user will be able to see all the values collected from the soil, which are stored in the server. On the other hand, Google Colab is a cloud-based environment, where programs can be uploaded and run without the use of any hardware device. The data acquired from the sensors is sent to the Blynk server for storing, and subsequently, via the API, the stored data is uploaded to Google Colab. Then the prediction from Google Colab is

again sent to the Blynk server. Apart from that, through the sensor-collected data, the system can easily determine the amount of fertilizer that should be applied for ideal crop production.

The NPK sensor takes the nutrient value in milligram per kilogram unit, which can be considered as an issue while understanding the amount. Therefore, to convert the unit to kilogram per hectare (1) can be used [13].

$$kg/ha = \frac{mg/kg \cdot \text{soil layer thickness} \cdot \text{bulk density}}{10} \quad (1)$$

Soil bulk density is the mass of soil per unit volume. Average bulk density of soil is between 1.0 to 1.4 g/cm³ so, 1.2 g/cm³ was used as an average. Again, soil layer thickness can vary based on land type but for Bangladesh the topsoil is (10-15) cm thick [14]. Thus, 12 cm was selected on an average for soil thickness. For the proposed system to work, an uninterrupted internet connection is required.

IV. SYSTEM IMPLEMENTATION

Implementation of the proposed soil monitoring with crop and fertilizer recommendation system contains various types of hardware, IoT-based systems, cloud computing, and user

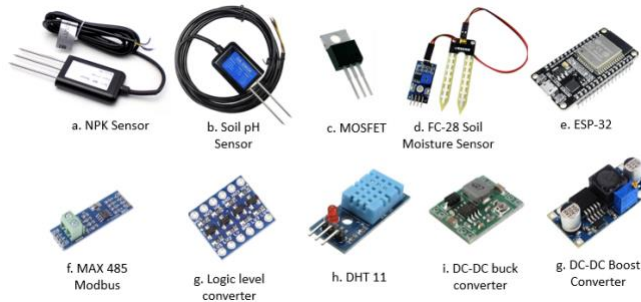


Fig. 2. Components used for the hardware development

interface device. For collecting all the required parameters from soil, various hardware components are used, as shown in Fig. 2.

The system uses ESP-32 with integrated Wi-Fi module for IoT integration and data processing. But, for simulation purposes the Arduino library has been used, as it is easily available and has the same functionality as ESP-32. Moreover, an NPK sensor is used for collecting nutrient values, i.e., amount of nitrogen, phosphorus, and potassium in the soil. Moreover, a soil pH sensor is used to determine the pH level of the soil, FC-28 is used for soil moisture level acquisition and DHT11 is used for sensing surrounding humidity and temperature.

A. Software simulation for data collection

In order, to evaluate the potential of the proposed system, a simulation was carried out before hardware implementation, shown in Fig 3. Through simulation, the functionality and validity of the proposed system can be observed.

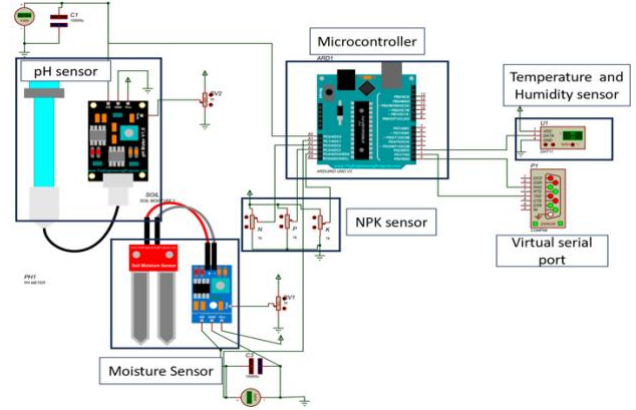


Fig. 3. Simulation of proposed system

For simulation purposes Proteus 8.11 is used because of its user-friendly environment for microcontroller-based simulation. Building a link between the simulation circuit inside Proteus and Blynk was made feasible using a virtual comport feature inside the proteus environment. This virtual serial port models the physical serial port in the simulation environment. The changed value of sensors in Proteus are transmitted to Blynk server with the help of this virtual feature of Proteus.

B. Hardware set up for data collection

Following a successful simulation of the proposed system, hardware implementation was executed. A portable system with all the sensors implemented in the microcontroller, along with a battery as a power source is developed. While testing the sensor data collection, a short-circuit problem was encountered. While placing the sensors next to each other, the probes of the sensors discharge current, leading to short-circuit issues and erroneous data acquisition.

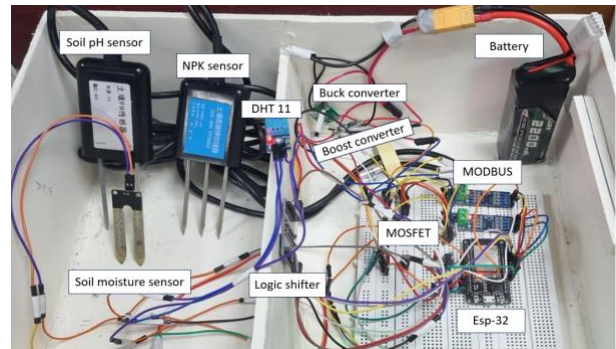


Fig. 4. Prototype of the proposed system

For gaining control over the operation of each sensor MOSFETs were used for switching one sensor at a time. As mentioned previously, to avail the portability features of the device, a 11.9 V battery was used and later converted to the required voltage for the sensors, utilizing buck and boost converter for powering up necessary components and thus, led

to successful hardware implementation. Fig. 4 demonstrates the real-life prototype of the proposed system. Additionally, a soil bed was constructed for hardware testing in the laboratory, and separate compartments for the same soil sample were constructed to prevent short-circuiting. The soil samples have been collected from three different districts of Bangladesh showed in Fig. 5, which are Barguna, Tangail and Khulna. The samples were preserved inside the laboratory and tested on a regular basis to observe the changes. Only the moisture level of the soil was impacted due to the preservation of soil samples. After the hardware implementation, Blynk IoT application is integrated to the ESP-32 of the system. The average value of all the parameters from the selected dataset is shown in Table I.

TABLE I. MEAN VALUES FOR ALL PARAMETERS FOR THE SELECTED CROPS

Parameters	Cotton	Jute	Lentil	Maize	Rice
Temperature (°C)	31.23	26.11	35.87	32.83	32.61
Humidity (%)	58.90	82.49	58.08	77.03	70.25
Soil pH	6.3	7.0	7.1	5.9	5.6
Nitrogen (mg/kg)	102.35	203.04	130	188.66	163.77
Phosphorus (mg/kg)	51.70	189.69	37.70	66.27	112.77
Potassium (mg/kg)	92.55	201.49	139.32	198.62	134.26

Afterwards the data collected through the prototype, the data are serially transferred through the Wi-Fi feature of ESP-32 to the Blynk server. To predict crops, a classification-



Fig. 5. Soil sample test with prototype in laboratory

based machine learning algorithm is used. Although there are several paid cloud computing platforms available, Blynk IoT and Google Colab are utilized for prototyping purposes. The data collected from the sensors are transmitted to the Blynk server for storage, and through the Application Programming Interface (API), the stored data is then sent to Google Colab in real ti

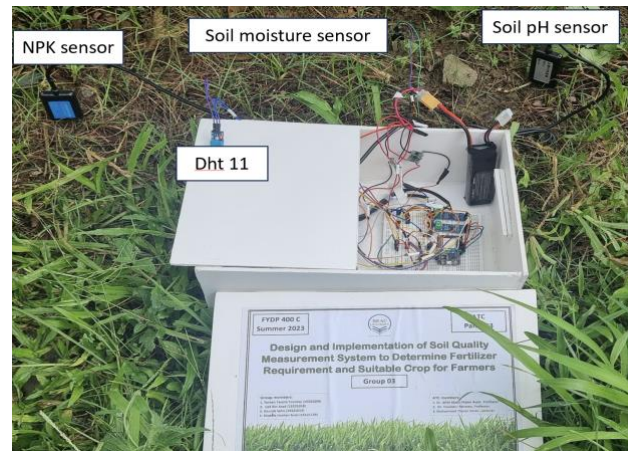


Fig. 6. Field test of the prototype

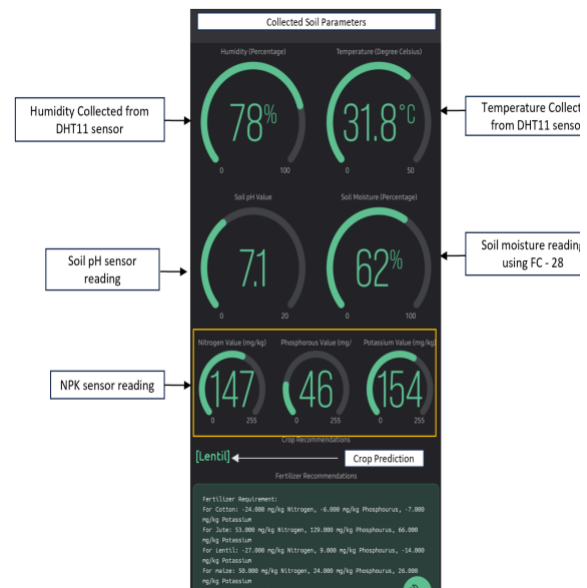


Fig. 7. Output in Blynk IoT mobile application interface

V. RESULT AND ANALYSIS

The implemented system can be segmented into two parts, where the initial part is data acquisition through all the sensors. Subsequently, there is a cloud-based decision-making system that was implemented with the help of ML models.

The process of collecting data in real life using the prototype is depicted in Fig. 6. Data from each sensor is collected sequentially with a 2-second delay and transmitted to the Blynk server via Wi-Fi. In order to guarantee accurate data collection using NPK and pH sensors, it is required that the soil samples have an adequate amount of moisture.

After a successful test of the prototype inside the laboratory, the system is also tested in real life by visiting a nearby field to measure the soil parameters and verifying the prediction and suggestions whether the soil is suitable for cultivation or not.

As the sensor probes were being inserted into the soil, the Blynk IoT mobile app began displaying all the soil parameters. The system required a minimum of 30 to 40 seconds to stabilize the value, after which it displayed the optimal prediction. Blynk, the mobile app based on IoT, features a straightforward interface and is highly suitable for data presentation. The Blynk interface in Fig. 7 displays the acquisition of sensor data, as well as the recommendations for crop and fertilizer.

Here, in the top part of the interface there is an indication of humidity in the left-hand side and temperature in the right-hand side, which is collected from DHT11 sensor. Underneath that in the left portion of the middle a pH reading of 7.1 is visible which indicated a neutral condition of soil and besides that 62 percent of humidity can be observed. In the third serial, the reading of nitrogen, phosphorus and potassium are shown serially in a row. These are the vital nutrients of the soil and have great impact in fertilizer amount calculations.

After the data being collected and stored in Blynk then comes the data integration in ML models. For performance analysis of the machine learning algorithm, the crop recommendation dataset was used hence, this dataset encompasses all the parameters of soil according to the requirements [12]. Multiple ML models are used to observe which provides better accuracy. A comparative table for analysis is shown in Table II.

TABLE II. COMPARISON OF DATASET USED IN SIMULATION AND DATASET COLLECTED USING PROTOTYPE

Sl.	Name of Algorithms	Test Accuracy
1.	Logistic Regression	81.33%
2.	Decision Tree	90.67%
3.	Random Forest	91.36%
4.	Support Vector Machine	88%

For a given algorithm, the confusion matrix indicates the total number of real and predicted labels. So, based on the test data from fertilizer recommendation dataset [12], a confusion matrix was generated using random forest algorithm as shown in Fig. 8. Here, a total of five crops which are jute, lentil, maize, cotton, and rice are selected for prototype testing purposes. From the confusion matrix, it can be observed that eight times our model predicted jute while it should have been rice. Similarly, four times the model predicted rice while it should have been jute. This issue is highly dependable on the accuracy of the model. A test case scenario of hardware and software-based system, conducted in lab, is presented in Table III.

True	Cotton	31	0	0	0	0
	Jute	0	27	0	0	4
	Lentil	0	0	26	0	0
	Maize	1	0	0	22	0
	Rice	0	8	0	0	31
	Predicted	Cotton	Jute	Lentil	Maize	Rice

Fig. 8. Confusion matrix for random forest

TABLE III. OUTPUT MATCHING PREDICTION FOR BOTH SOFTWARE AND HARDWARE BASED SYSTEM

Param	SW	HW	SW	HW	SW	HW
N	190	190	125	125	156	156
P	65	65	37	37	116	116
K	197	197	132	132	137	137
Temp	23	22	24.5	22	27	22
Moist	37	37	52	52	75	75
pH	7.8	7.8	7	7	6.7	6.7
Hum	63	70	68	70	75	70
Prediction	maize	maize	lentil	lentil	rice	jute
Status	Matched		Matched		Not matched	

In Table III, the left column indicates all the parameters like nitrogen (N), phosphorus (P), potassium (K), temperature (Temp), moisture (Moist), pH, humidity (Hum) values which is implied in both software and hardware system. Due to lab environment the temperature and humidity remained similar in hardware testing. The prediction between rice and jute seems inconsistent from the analysis and other than that all the prediction matches for remaining cases. Furthermore, moisture is not included in the dataset as this parameter is additionally taken to measure the amount of water threshold in the soil for individual crops. For instance, the presence of moisture in paddy and cotton cultivating field will not be same. A comparison among the three major nutrients for the soil is shown in Fig. 9.

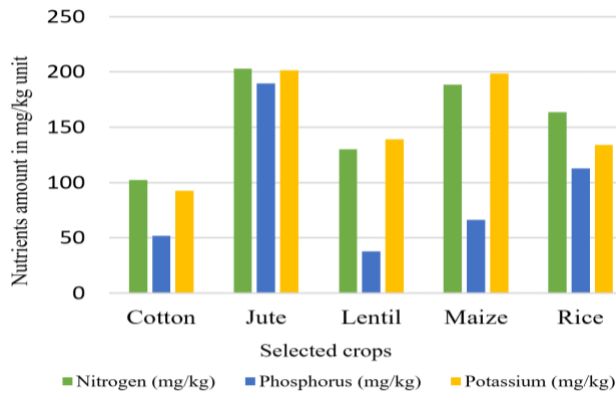


Fig. 9. Comparison between the major nutrients of the selected crops

Observing Table IV, it can be compared that for both soil samples, which are from Tangail and outdoor field, there remains a similarity in values. For both cases the prediction is “Lentil” and this shows the validity of the system. In the indoor environment, the temperature was the only parameter that could not be replicated. But due to that one factor the output did not change drastically since the features like N, P, K has more dependency on the system in comparison to other parameters.

TABLE IV. COMPARITIVE DATA OF DIFFERENT SOIL SAMPLE IN BOTH INDOOR AND OUTDOOR CASE

Parameters	Laboratory soil sample test			Outdoor Test
	Barguna	Tangail	Khulna	Outdoor field
Temperature (°C)	22	22	22	31.8
Humidity (%)	70	70	70	78
Moisture (%)	37	52	75	62
Soil pH	7.8	7	7.4	7.1
Nitrogen (mg/kg)	190	125	224	147
Phosphorus (mg/kg)	65	37	189	46
Potassium (mg/kg)	197	132	228	154
Crop Prediction	Maize	Lentil	Jute	Lentil

The fertilizer suggestions for different crops are shown in the Blynk interface in Fig. 10. It is natural that the excess nutrients present in the soil cannot be eliminated, but adding the required amount for making the soil optimal for selected crop production is possible. Based on the fertilizer requirement shown in Blynk app, the application amount of fertilizer is presented in Table V. The table shows the amount of fertilizer to be applied in kilogram per hectare.

Table V represents the applicable amount of major fertilizer which contains nitrogen, phosphorus, and potassium. Generally, Urea contains 46 percent nitrogen. Again, Triple Super Phosphate (TSP) contains 20 percent phosphorus.

Meanwhile, Muriate of Potash (MOP) contains 60 - 62 percent potassium. Based on the amount of these major nutrients present in soil, Table V is made along with the reference to fertilizer recommendation guide 2018 [15]. With the increasing amount in milligram per kilogram the fertilizer amount with the unit of kilogram per hectare increases linearly.

TABLE V. APPLICABLE AMOUNT OF FERTILIZERS CHART

Fertilizer Requirement by the system	Amount of fertilizer		
	Urea	TSP	MOP
Less than 0 mg/kg	N/A	N/A	N/A
10 mg/kg	7.8 kg/ha	18 kg/ha	6 kg/ha
20 mg/kg	15.6 kg/ha	36 kg/ha	12 kg/ha
30 mg/kg	23.4 kg/ha	54 kg/ha	18 kg/ha
40 mg/kg	31.2 kg/ha	72 kg/ha	24 kg/ha
50 mg/kg	39 kg/ha	90 kg/ha	30 kg/ha
60 mg/kg	46.8 kg/ha	108 kg/ha	36 kg/ha
70 mg/kg	54.6 kg/ha	126 kg/ha	42 kg/ha
80 mg/kg	62.4 kg/ha	144 kg/ha	48 kg/ha
90 mg/kg	70.2 kg/ha	162 kg/ha	54 kg/ha
100 mg/kg	78 kg/ha	180 kg/ha	60 kg/ha

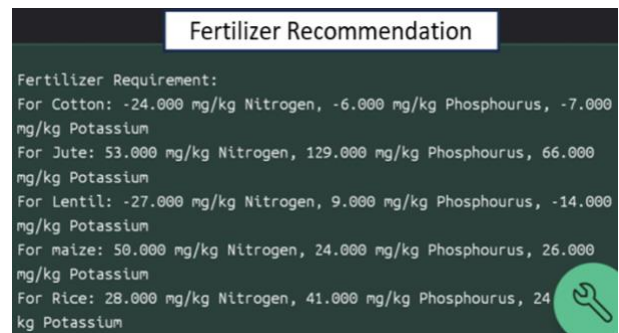


Fig. 10. Fertilizer suggestion in Blynk IoT mobile Application Interface

There are organizations initiated by the government that conduct soil tests and analyse the quantity of beneficial nutrients in the soil. The technical data are not that useful to the farmers, that is why a smart soil monitoring system is proposed which gives the user easy access to their soil data in the palm of their hand. There are separate devices available in the market which can measure the soil parameters, but those data cannot be utilized for further analysis. The system proposed and implemented in this research paper has the ability of utilizing those collected data of soil for further

prediction of suitable crops for a particular land along with suggesting the amount of optimal fertilizer

VI. CONCLUSION AND FUTURE WORK

In conclusion, an IoT based smart soil monitoring system has been developed along with suitable crop and fertilizer recommendation. The added mobile interface made it more user friendly. It is developed for the farmers so that they get to know the soil quality of their land. It is obvious for the farmers not to have a deep analysis of the land they are cultivating, but with the help of modern technology, it is now possible. There are various meters available for measuring the soil parameters, but the system can, not only collect the data and show the soil parameters if anyone wants to understand the soil nutrient level, but also help the user with recommendation features in crop selection and fertilizer suggestions. The system can be more optimized by collecting more crop data. Moreover, it can be integrated for smart farming and used for large projects by fixing the device at a specific place while linking multiple similar devices for seamless data analysis. For future improvement a data storing system can be adjoined. Implementing these additional features will help in more accurate recommendations of different crop variety.

ACKNOWLEDGMENT

The authors acknowledge the Control & Applications Research Centre (CARC) and EEE Department of BRAC University for guidance from the beginning to end.

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